大一

繼續精進學業，持續挑戰

持續保持優異的學業成績

醫學系系排名 上學期 23 下學期 11 總學年 12

大量參與運動社團，強身健體(系桌、系籃)

院際盃桌球錦標賽冠軍



進入實驗室，學習實驗操作技術

海報論文

Development of Peptide Blockers for Preventing OSCC Recurrence through Disrupting Nerve-Tumor Communication

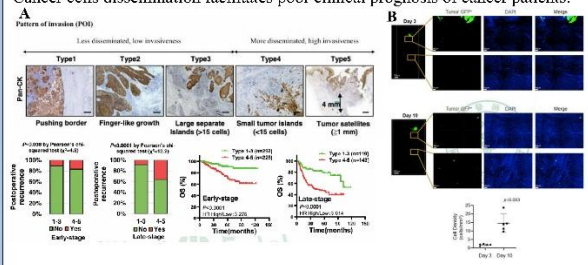
Chang-Hung Hsieh[1], Yen-Tang Chan[1], Yun-Ting Yeh[1], Yung-Chun Chang[1], Heng-Hsiung Wu[2]

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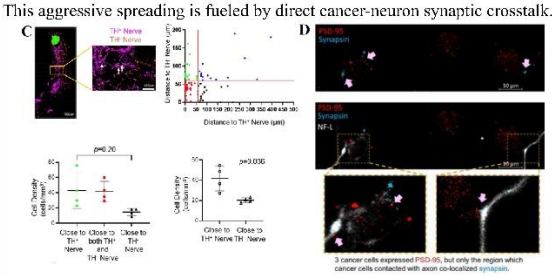
Laboratory of Cancer Neuroscience

Introduction

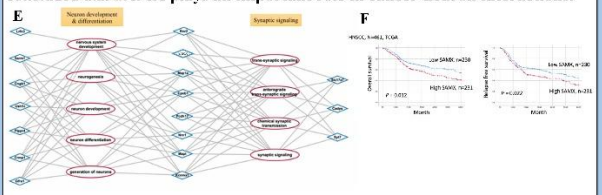
Cancer cells dissemination facilitates poor clinical prognosis of cancer patients.



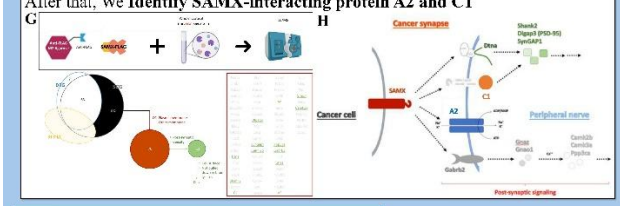
This aggressive spreading is fueled by direct cancer-neuron synaptic crosstalk.



By analyzing genes related to neuron development and synaptic signaling, we concluded that **SAMX plays an important role in cancer-neuron interactions.**



After that, We **Identify SAMX-interacting protein A2 and C1**



Aims

After targeting SAMX as the key mediator of cancer-communication, we aim to:

1. Identify the SAMX–ligand binding interface by Alphafold3
2. Design and prioritize SAMX-targeting peptide blockers by Bindcraft
3. Validate peptide blockers in a tumor–nerve co-culture model

References

- N. Vigneswaran and M. D. Williams, "Epidemiological Trends in Head and Neck Cancer and Aids in Diagnosis," Oral Maxillofac. Surg. Clin. N. Am., vol. 26, no. 2, pp. 123–137, May 2014, doi: 10.1016/j.coms.2013.01.001.
- Chen, Y.-T. (2023). *To investigate the roles of nerves in head and neck squamous cell carcinoma* [Master's thesis, China Medical University].
- Chen, K.-T. (2024). *The roles of Neurexin 2 in the interactions between oral cancer cells and nerves* [Master's thesis, China Medical University].
- "Using AlphaFold-Multimer to discover interaction protein-protein interactions - Humura - 2024 - The Plant Journal - Wiley Online Library." Accessed: Jan. 12, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/full/10.1111/tpj.16669

Methodology

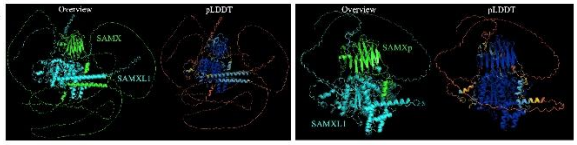
AlphaFold3 model SAMX interactions with defined ligands (SAMXL1/2) and candidate ligands (A2/C1) to identify shared binding interfaces

BindCraft design peptide blockers targeting the predicted SAMX interface prioritize structural confidence, hydrophilicity and stability

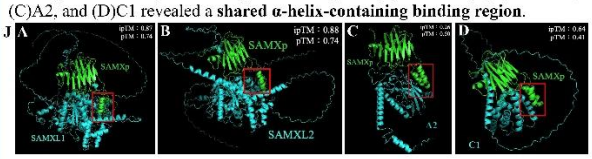
SCG-derived peripheral neuron–OSCC co-culture test

Preliminary Results

SAMX structure shows low confidence, so we remove the instability region to generate a more stable core structure termed **SAMXp, which substantially improved the confidence of interaction predictions.**



Subsequent modeling of SAMXp with its ligands (A)SAMXL1, (B)SAMXL2, (C)A2, and (D)C1 revealed a **shared α-helix-containing binding region.**

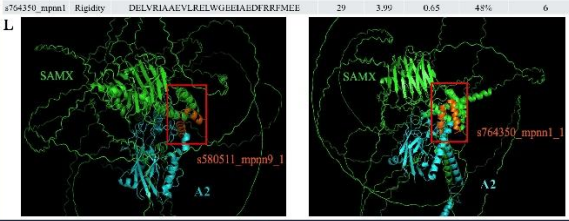


Targeting this α-helix region, we **develop two lead peptides based on structural confidence、hydrophilicity and stability.**

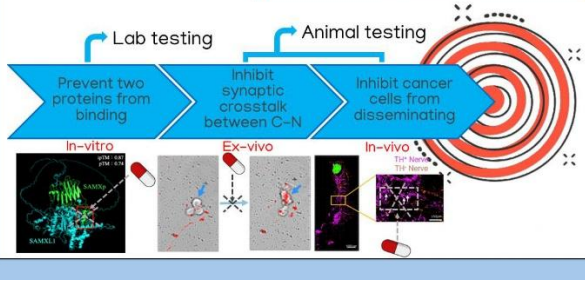
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Peptide	Mode	Sequence	Residues	Isoelectric point	Average hydrophilicity	Ratio of hydrophilic residues	Blocking C1 residues
s580511_mpnn9	Flexible	SEEEIKAKMDDIWQKNPANFFDDLIAWLS	29	3.87	0.26	52%	9
s764350_mpnn1	Rigidity	DELVRIAAEVLRELWGEEIAEDFRRFMEE	29	3.99	0.65	48%	6

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Future



海報論文特優

【恭賀】115年大專生科學研習營成果發表會論文競賽得獎名單

公告日期: 2026-11-04
類別: 最新消息
115年大專生科學研習營成果發表會於10月31日舉行，參賽同學透過口頭及海報發表，展現研究成果。恭喜以下同學於論文競賽中榮獲佳績！

一、口頭論文

獎項	姓名	獎金（元）
特優	詹O堂	5,000
優等	簡O煌	3,500
優等	謝O璇	3,500
佳作	張O平	2,500
佳作	葉O穎	2,500
人氣獎	謝O哲	1,500
人氣獎	廖O英	1,500
人氣獎	葉O穎	1,500

二、海報論文

獎項	姓名	獎金（元）
特優	謝O宏	4,000

金頭腦科研計畫簡報競賽 佳作



赴泰國交換，培養國際視野與英文能力



